DBA-MA: International Framework

Expanding the Design Basis for Nuclear Power Plants

To Address Military Action Against Nuclear Facilities

An International Framework

Based on Observations from the Russia–Ukraine Conflict
and the IAEA's Findings at the Zaporizhzhia Nuclear Power Plant

Version 2.2

October 2026

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T. Roxey, "Expanding the Design Basis for Nuclear Power Plants to Address Military Action Against Nuclear Facilities: An International Framework," Version 2.2, Eclectic Technologies, October 2026.
DOI (v2.2): https://doi.org/10.5281/zenodo.23420400. Supersedes v2.0, https://doi.org/10.5281/zenodo.22944855; v2.1 was not deposited.
Current version: https://scubanuke.github.io/publications/dba-ma/L1_DBA-MA_IAEA_Framework.pdf
Prepared for the International Nuclear Safety and Security Community

This document serves as the parent framework for the DBA-MA series.

National implementation documents for specific regulatory jurisdictions are developed separately.

1. Purpose and Proposal

This document proposes developing an international framework to incorporate military action against nuclear facilities into the design basis of nuclear power plants. The events observed during the Russia–Ukraine conflict, beginning with the full-scale invasion of February 24, 2022, demonstrated that military operations against operating nuclear power plants are no longer a theoretical possibility but an observed reality. The international nuclear safety framework must account for this reality.

The framework presented here is intended to be jurisdiction-neutral. It establishes the engineering principles, safety objectives, and general coordination architecture that any national regulatory authority can adopt and implement within its own legal and institutional context. It is framed in the vocabulary and standards architecture of the International Atomic Energy Agency, drawing on the IAEA Safety Fundamentals (SF-1), the Safety of Nuclear Power Plants: Design requirements (SSR-2/1), and the Nuclear Security Series, as well as the IAEA's own operational findings from its sustained presence at the Zaporizhzhia Nuclear Power Plant.

This proposal does not suggest that commercial nuclear operators should be expected to repel military forces. Rather, it argues that the engineering design and operational posture of nuclear facilities should be robust enough to maintain essential safety functions under the conditions that military action creates—sustained loss of external support systems, physical damage from explosive ordnance, degraded staffing, disrupted supply chains, and the simultaneous challenge of multiple safety functions over extended durations.

The framework is written for nuclear power plants, where the evidence of Section 2 was gathered. Its principles apply by analogy to other nuclear installations, including research reactors and the fuel cycle facilities addressed by IAEA SSR-4. Each facility class differs in its governing consequences: at an enrichment or conversion plant, for example, the release of uranium hexafluoride and the criticality of rearranged fissile material take the place of decay heat removal. An instrument applying this framework to another facility class therefore states that class’s governing consequences before applying the principles.

The document is structured as the parent framework in a series designated DBA-MA (Design Basis Accident—Military Action). National implementation documents, addressing the specific regulatory frameworks, authority structures, and coordination mechanisms of individual member states, are developed as child documents under this framework. The first such child document addresses the United States regulatory context; additional national implementations are anticipated for European member states with particular exposure to the demonstrated threat.

2. The Ukraine Precedent: Operational Evidence for Framework Revision

The Russia–Ukraine conflict produced the first sustained military operations against operating nuclear power plants in the history of commercial nuclear energy. The evidence from these operations, documented extensively by the IAEA through direct on-site presence and published reporting, constitutes the empirical foundation for this framework.

2.1 The Chornobyl Seizure

Russian forces seized the Chornobyl Exclusion Zone on the first day of the invasion. Although not an operating power reactor, the facility remained a significant radiological concern due to the spent-fuel storage installations and the containment structure over the damaged Unit 4 reactor. The seizure demonstrated that military forces will occupy nuclear facilities as part of broader operational objectives without regard to nuclear safety considerations. Staff were held under coercive conditions, communications were disrupted, and normal safety and security routines were compromised for the duration of the occupation.

2.2 The Zaporizhzhia Nuclear Power Plant

The seizure and prolonged occupation of the Zaporizhzhia Nuclear Power Plant (ZNPP) presented the more consequential scenario. ZNPP, a six-unit VVER-1000 pressurized water reactor complex with a total generating capacity of approximately 5,700 MWe, was the largest nuclear power plant in Europe at the time of its seizure. During the period of active conflict around the facility and the subsequent occupation, ZNPP faced conditions that systematically challenged every assumption underlying the existing design-basis framework.

The IAEA, through its Director General and successive on-site inspection and support missions, reported that all seven pillars of nuclear safety and security were compromised at various points during the occupation. These pillars—physical integrity, safety and security systems, operating staff, off-site power supply, logistics and supply chain, radiation monitoring, and communications—represent the IAEA's own framework for the essential conditions that must be maintained at any nuclear facility. The Director General’s second summary report, covering April to September 2022, records that the shelling of 5 and 6 August 2022 alone compromised all seven pillars at once. That finding, the simultaneous compromise of every pillar, is the single most important piece of evidence supporting this proposal.

The specific conditions observed at ZNPP included: repeated loss of off-site power from all available transmission lines, forcing reliance on emergency diesel generators and creating conditions analogous to a station blackout; physical damage to structures and systems from shelling and explosive munitions; coercive conditions imposed on operating staff, including armed military presence in the control room, detention of personnel, and interference with shift rotations; complete disruption of the normal supply chain for fuel, spare parts, and consumables; severing of normal communications between the plant staff, the national regulator (SNRIU), and the IAEA; and military emplacement of equipment and ordnance on the plant site in a manner inconsistent with nuclear safety requirements.

2.3 The Significance for the International Safety Framework

The significance of these events for the international design basis framework lies not in their unprecedented nature taken individually—loss of power, loss of cooling, staffing disruptions, and supply chain failures are all addressed individually in existing safety standards—but in their simultaneous occurrence, their deliberate causation, their indefinite duration, and the systematic degradation of every external support system on which existing safety analyses depend.

Existing design-basis events assume that initiating events are independent, that a defined progression and response follows them, and that external support—off-site power, emergency response organizations, supply chains, and regulatory oversight—remains available to support recovery. Military action against a nuclear facility violates all of these assumptions simultaneously. It creates a correlated failure environment in which multiple safety functions are challenged simultaneously by a persistent, adaptive adversary whose actions cannot be modeled by probabilistic methods used for natural hazards and equipment failures.

2.4 Uncrewed Aerial Systems: A Distinct Dimension of the Ukraine Precedent

While the occupation of ZNPP and Chornobyl demonstrated the conditions created by conventional military seizure, the broader Russia–Ukraine conflict has simultaneously demonstrated a distinct and rapidly evolving mode of military action against energy infrastructure: sustained operations by uncrewed aerial systems. Russia's campaign against Ukrainian generation, transmission, and distribution assets has employed UAS at a scale unprecedented in the history of warfare against civilian infrastructure. This campaign has included waves of one-way attack drones (loitering munitions) numbering dozens to hundreds per strike; first-person-view tactical drones directed against specific infrastructure components, vehicles, and personnel; reconnaissance UAS providing continuous pre-strike surveillance and post-strike battle damage assessment; coordinated operations combining UAS with cruise and ballistic missiles to saturate air defenses; and, increasingly, AI-enabled autonomous platforms demonstrating jamming resistance, autonomous navigation, and strike effectiveness rates reported in open-source accounts as high as eighty percent (Kyiv Post, October 2024).

The UAS threat creates conditions materially distinct from those addressed by the ZNPP precedent alone. UAS enable sustained-tempo attacks over weeks to months at a fraction of the cost of conventional missile campaigns, imposing a destruction-and-re-destruction cycle in which replacement equipment is targeted because its location is known from persistent surveillance. They provide the adversary with continuous situational awareness of the facility and its operations, enabling adaptive, precision targeting of identified vulnerabilities. They enable compound attack modalities—for example, cyber-enabled tripping of reactor output breakers combined with UAS destruction of the switchyard to prevent power restoration—that can produce station blackout conditions of indefinite duration without directly striking the reactor. Moreover, the rapid integration of artificial intelligence into drone operations means countermeasures that rely on disrupting operator control links are becoming progressively less

effective. The UAS dimension of the Ukraine precedent requires dedicated treatment within the DBA-MA framework; the companion instruments DBA-UxS and DBA-MA-UxS-NA elaborate on this analysis.

2.5 The Threat Categories

The evidence above resolves into six categories of military action, distinguished by their mechanism of effect, which the DBA-MA series uses as its threat taxonomy. The first is seizure and occupation of the facility, as at Chornobyl and ZNPP. The second is bombardment by explosive ordnance, whether artillery, rockets, missiles or aerial munitions. The third is attack by uncrewed aerial systems, treated as its own category because of its precision, its tempo and its surveillance-to-strike cycle (Section 2.4). The fourth is cyber attack on plant digital systems, alone or combined with kinetic action. The fifth is denial of external support: off-site power, supply chains, communications, and outside technical and regulatory support. The sixth is electromagnetic attack by nuclear electromagnetic pulse or high-power microwave (Section 5.7).

An actual attack will usually combine several categories, and it is the combination, with its correlated failures, that distinguishes military action from the independent initiating events of the existing design basis. The threat characterization for each facility determines which categories are credible at its site; the national implementation documents and the cross-cutting siting methodology, DBA-MA-SITE, govern that determination.

3. The Gap in Existing International Safety Standards

3.1 The IAEA Safety Standards Framework

The IAEA safety standards are organized in a hierarchy: Safety Fundamentals (SF-1), which establishes the fundamental safety objective and ten safety principles; Safety Requirements, which set out requirements to be met for the protection of people and the environment; and Safety Guides, which provide recommendations and guidance on meeting the requirements. For nuclear power plant design, the primary requirements document is SSR-2/1, Safety of Nuclear Power Plants: Design, which establishes the design requirements, including the specification of postulated initiating events, design-basis accidents, and design-extension conditions.

SSR-2/1 requires that the design basis for a nuclear power plant include a comprehensive set of postulated initiating events, encompassing internal events (equipment failures, human errors), internal hazards (fires, floods, missiles), and external hazards (seismic events, extreme weather, aircraft crash). Design extension conditions (DECs) extend the analysis beyond the design basis to address less frequent but more severe scenarios, including severe accidents with core damage. The DEC concept, introduced in the 2012 revision of SSR-2/1, significantly expanded the analytical scope following the Fukushima Daiichi accident.

However, neither the design basis events nor the design extension conditions currently account for the deliberate, sustained, and correlated disruption of safety functions resulting from military action. The external hazards considered in SSR-2/1—including the provisions for aircraft crash analysis introduced or strengthened after September 11, 2001—address discrete events with defined physical parameters. Military action differs categorically: it is sustained rather than transient, adaptive rather than static, and targets multiple safety functions and all external support systems simultaneously.

3.2 The Nuclear Security Framework

The IAEA Nuclear Security Series addresses threats to nuclear facilities using the Design Basis Threat (DBT) concept. The DBT characterizes the attributes and capabilities of potential adversaries—insiders, outsiders, or combinations thereof—that the physical protection system must defend against. National DBTs are developed by member states based on their own threat assessments and are typically classified.

The DBT framework, however, explicitly addresses subnational threats from criminal, terrorist, or sabotage actors. It does not extend to military forces operating under a nation-state’s authority. This exclusion reflects the long-standing assumption, shared by virtually all national regulatory frameworks, that the defense of nuclear facilities against military attack is a sovereign military responsibility rather than a facility-design requirement. The events in Ukraine have demonstrated that this assumption—however reasonable it appeared before February 2022—does not hold under all conditions. National military defense may fail to prevent the seizure of a nuclear facility. When it does, the facility's own engineering design becomes the last line of defense for radiological safety.

International humanitarian law was intended to make this question unnecessary. Article 56 of Additional Protocol I to the Geneva Conventions (1977) protects works and installations containing dangerous forces, and names nuclear electrical generating stations among them, from being made the object of attack where the attack may release dangerous forces and cause severe losses among the civilian population. That protection has limits a design basis cannot ignore. Not every state is party to the Protocol; for a nuclear generating station the special protection ceases if the station provides electric power in regular, significant and direct support of military operations and attack is the only feasible way to end that support; and in every case the protection depends on the conduct of the attacker. After the seizure of ZNPP, the IAEA Director General set out five principles for the plant’s protection to the United Nations Security Council on 30 May 2023, the first of them that there should be no attack of any kind from or against the plant. The law and the principles remain essential. The Zaporizhzhia experience shows that a facility’s safety cannot be made to depend on them.

3.3 The Gap Between Safety and Security

The military action scenario falls between the IAEA's safety standards and its security guidance. The safety framework addresses the engineering response to events but excludes deliberate military action from its event catalog. The security framework addresses deliberate adversarial action but limits its scope to sub-national threats. Military action by a nation-state falls outside both frameworks.

This document argues that the appropriate response is to address military action primarily through the safety standards framework—specifically, by establishing the conditions created by military action as a new category of design extension condition, analysed with the discipline of the design basis (Section 8.1)—and to draw on the security framework for threat characterization inputs. The rationale is straightforward: the engineering design principles needed to survive military action conditions are safety engineering principles (redundancy, independence, passive safety, extended coping times), not security measures (guards, barriers, detection systems). The facility cannot defeat the military threat; it must survive the conditions the threat creates. That is a safety design problem.

3.4 The Geopolitical Stability Dimension of Siting Assessment

Existing IAEA siting criteria — principally NS-R-3 (Revision 1) and its successor SSR-1 — address the external hazard environment of a proposed nuclear facility site: seismic activity, flooding, extreme meteorological events, aircraft crash frequency, and the proximity of hazardous industrial activities. Geopolitical stability is not among the established siting criteria. The implicit assumption has been that states assess the political and military risk environment as part of the sovereign decision to pursue nuclear power, and that the regulatory framework governing facility design need not engage with it. The Zaporizhzhia precedent requires revisiting this assumption. A facility sited and licensed under stable political conditions may find itself in a contested military environment during its operating life. Geopolitical stability is not a fixed characteristic of a site; it is a time-varying condition whose deterioration can place a facility in precisely the circumstances this framework addresses. The DBA-MA framework introduces a geopolitical stability dimension to siting assessment that is absent from existing criteria: a proposed site’s location relative to contested borders, the stability of the host state’s political environment, the proximity of allied and adversary military assets, and the strategic value of the facility as a potential military objective are all factors that bear on the probability and consequence of military action against the facility across its projected operating life. National child documents should address the mechanism by which this dimension is incorporated into their jurisdiction’s siting review process. The cross-cutting instrument DBA-MA-SITE, the Geopolitical Siting Assessment Methodology, develops this dimension into a working method.

This dimension is particularly consequential for small modular reactor (SMR) deployment. SMRs are being developed and promoted specifically for deployment in regions where large conventional nuclear plants are impractical — including remote areas, developing nations, and locations at the margins of grid infrastructure. Many of the environments where SMR

deployment is most commercially attractive are also environments of elevated geopolitical complexity: regions with contested sovereignty, proximity to active or latent military conflicts, or dependence on the political relationships of a small number of state actors. SMR proponents have argued that the smaller source term of an SMR — the reduced inventory of radioactive material relative to a conventional large reactor — justifies relaxation of siting criteria, including increased proximity to population centers and reduced emergency planning zone requirements. That argument is made within the existing siting framework, which does not include geopolitical stability as a variable. A source-term-based relaxation of siting criteria that reduces emergency planning requirements may be analytically defensible in a stable geopolitical environment; it requires independent evaluation of the DBA-MA threat categories (Section 2.5) for the proposed site before it can be accepted as adequate in a geopolitically complex one. The DBA-MA framework does not prohibit SMR deployment in any location. It requires that the military-action threat category evaluation be performed independently of the source-term analysis, and that source-term-based arguments for siting relaxation not be accepted as sufficient without that evaluation.

4. Alignment with IAEA Fundamental Safety Principles

The proposal to incorporate military action into the design basis is not a departure from the IAEA's safety philosophy but a logical extension of principles already established in SF-1. Several of the ten fundamental safety principles directly support this framework.

Principle 3, Leadership and Management for Safety, requires that safety leadership be demonstrated at the highest levels and that an effective management system be established and sustained. The management system for a nuclear facility operating under conditions of military threat requires planning, defined procedures, and trained personnel—none of which can be improvised during a crisis. The absence of a design basis framework for military action means there is no systematic management preparation.

Principle 5, Optimization of Protection, requires that protection be optimized to provide the highest level of safety reasonably achievable. In a world where military action against nuclear facilities has been demonstrated, optimized protection must account for this possibility, particularly for facilities in regions of elevated geopolitical risk.

Principle 8, Prevention of Accidents, requires that all practical efforts be made to prevent nuclear accidents. The military action scenario creates conditions that allow multiple existing accident sequences to be initiated simultaneously. Preventing these accidents requires design provisions that account for the initiating conditions, which the current framework does not address.

Principle 9, Emergency Preparedness and Response, requires that arrangements be made for emergency preparedness and response for nuclear facilities. The military action scenario fundamentally disrupts the assumptions underlying existing emergency preparedness: that off-site emergency organizations can access the facility and its environs, that evacuation routes are

available, that communications function, and that the event has a defined progression toward resolution. Emergency preparedness for military action requires fundamentally different planning assumptions.

The IAEA's seven pillars of nuclear safety and security—articulated by Director General Grossi specifically in the context of the Zaporizhzhia situation—represent an institutional acknowledgment that the conditions created by military action are relevant to nuclear safety. This framework proposes to give those pillars engineering substance by translating the observation that military action compromises all seven pillars into design requirements that preserve safety functions when any pillar is compromised.

5. Engineering Design Principles for Military Action Resilience

The engineering response to the military action design basis builds upon the safety design principles established in SSR-2/1—defense in depth, redundancy, diversity, independence, fail-safe design, and the single-failure criterion—but applies them to a threat profile that existing standards do not contemplate. The following seven principles define the engineering requirements specific to the military action scenario.

Cyber attack is one of the six threat categories (Section 2.5) but is not a separate design principle in this framework. The prevention and detection of cyber attack on plant digital systems are governed by each state’s computer security regime, for which IAEA Nuclear Security Series No. 42-G provides the international guidance. What this framework adds is the requirement that the plant survive the effects of a successful cyber attack, including one coordinated with kinetic action as described in Section 2.4. Local sovereignty of safety functions, extended autonomous operation and graceful degradation apply to the loss or corruption of digital systems exactly as they apply to the loss of power or of staff.

5.1 Local Sovereignty of Safety Functions

Safety-critical systems must be capable of maintaining their protective function without dependence on any external system—such as off-site power, supply chains, communications, external technical support, or regulatory oversight. This principle extends the existing SSR-2/1 requirements for independence and redundancy to their logical conclusion under military action conditions: the facility must be able to maintain safe conditions using only the resources physically present at the site and under the control of the operating staff.

Local sovereignty does not mean that external support is never expected; rather, the design does not depend on external support to maintain fundamental safety functions during military action. This is analogous to, but more demanding than, the station blackout provisions that require the capability to operate for defined durations without off-site power. Under the military action scenario, the loss of external support is indefinite and encompasses all external systems, not only the electrical grid.

5.2 Extended Autonomous Operation

Systems designed for military action resilience must be capable of autonomous operation for durations far exceeding those contemplated by existing design-basis events or design-extension conditions. While existing standards address coping times measured in hours (for station blackout) or days (for some beyond-design-basis scenarios), military occupation in Ukraine has persisted for months or years. The design must account for this timescale, not by requiring months' worth of stored consumables, but by prioritizing passive safety features, natural-circulation cooling, and systems that can be sustained with locally available resources over extended periods.

5.3 Hardening Against Intentional Destructive Force

The existing external events analysis addresses natural phenomena and, following post-September 11, 2001, revisions, accidental or deliberate aircraft impact. The military action design basis must address the intentional application of destructive force by conventional military weapons, including direct and indirect fire from artillery and rocket systems, aerial munitions, and explosive ordnance of types observed in the conflict in Ukraine. The design response is not to fortify the facility against all military weapons—which is neither practical nor the purpose of this framework—but to ensure that safety-critical structures, systems, and components are protected or configured to maintain their function under the realistic range of damage that conventional military operations in the vicinity of a nuclear facility would produce.

Within this general hardening requirement, the threat posed by uncrewed aerial systems requires specific attention. The Russia–Ukraine conflict has demonstrated that UAS—including one-way attack drones, first-person-view tactical platforms, and autonomous AI-enabled systems—have become a dominant instrument of military operations against energy infrastructure. UAS present threat characteristics that differ materially from conventional artillery and missiles: they enable precision targeting of specific components rather than area bombardment; they can sustain a tempo of attacks over weeks to months at low cost to the adversary; they impose cumulative damage through repeated strikes at the same location; and they are preceded by persistent surveillance that enables the adversary to identify and target specific vulnerabilities. The hardening analysis must therefore address not only the blast and fragmentation effects of single strikes but the cumulative degradation of protective structures under repeated loading, the vulnerability of individual components (cable trays, valve operators, instrument connections) to small precision-guided munitions, and the protection of safety-significant systems whose locations become known through UAS reconnaissance. The companion instruments DBA-UxS and DBA-MA-UxS-NA develop these requirements in detail.

5.4 Degraded-Staffing Operability

The design must account for the possibility that normal staffing levels, expertise, shift rotation schedules, and working conditions may not be available. The Ukraine experience demonstrated that staff may be held under coercive conditions, prevented from rotating, denied access to rest facilities, and subjected to sustained psychological stress. The design response is to drive toward greater automatic actuation of safety-critical functions by hard-wired or qualified deterministic means, simplified operational procedures that can be executed under stress, and clear decision hierarchies that function when normal management structures are disrupted.

5.5 Graceful Degradation

Given the sustained and escalating nature of military action, the design should provide for graceful degradation of capability rather than cliff-edge failure. Systems should be designed so that progressive loss of capability results in progressively reduced but still safe plant conditions, rather than sudden transitions to uncontrolled states. This principle is already present in the defense-in-depth philosophy of SSR-2/1. Still, it must be extended to account for the deliberate, sequential degradation of multiple safety barriers that military action can produce.

5.6 Concealment and Signature Reduction

The persistent surveillance capability provided by reconnaissance UAS introduces a threat dimension that the five principles above do not address. An adversary conducting UAS surveillance over a nuclear facility can develop detailed, continuously updated knowledge of the facility's layout, the locations of safety-significant equipment, backup systems, consumable storage, and personnel shelters. This surveillance-to-strike cycle—in which reconnaissance UAS identify targets that strike UAS subsequently attack—has been a defining characteristic of modern drone warfare as observed in the Ukraine conflict. Replacement equipment and repair efforts are targeted because their locations are observed during installation.

The engineering response is to incorporate concealment and signature reduction into the design of safety-significant systems. Equipment that cannot be hardened to survive UAS strikes should, where practicable, be concealed from aerial observation through below-grade installation, the use of natural or constructed visual barriers, or camouflage. The thermal and electromagnetic signatures of operating equipment—visible to infrared and electronic surveillance sensors carried by UAS—should be assessed and, where practicable, reduced. Concealment does not replace hardening; it complements it by degrading the adversary's ability to identify and locate targets, thereby reducing the effectiveness of precision-guided strikes and increasing the number of UAS the adversary must expend to achieve a safety-significant effect. This principle extends the existing physical protection framework into a domain—counter-surveillance—that has not previously been within the scope of nuclear safety design.

5.7 Electromagnetic Pulse and High-Power Microwave

Military action against nuclear facilities may employ electromagnetic weapons — including nuclear-generated electromagnetic pulse (EMP) and directed-energy high-power microwave (HPM) devices — that threaten digital instrumentation and control systems through mechanisms distinct from kinetic damage. EMP produces a broad-spectrum, high-intensity electromagnetic transient capable of damaging or destroying electronic components across large areas without requiring line-of-sight or physical proximity to the target. HPM devices, which can be UAS-delivered, produce directed, high-intensity microwave energy capable of disabling or destroying digital electronics in targeted systems through soft-kill or hard-kill effects at standoff distances consistent with other UAS threat categories. Both threat types are characterized by their electromagnetic rather than kinetic mechanism of effect and by their particular effectiveness against modern digital I&C systems, which offer less inherent electromagnetic hardening than the analog relay-based protection systems they replaced.

The asymmetry between analog and digital plant I&C under this threat is a design consideration requiring explicit attention. Relay-based protection systems — characteristic of older plant designs — are substantially more resilient to EMP and HPM effects than solid-state digital systems, despite offering lower functional capability in other respects. Modern digital I&C upgrades may therefore reduce a facility's electromagnetic resilience even as they improve its operational and safety capabilities. This trade-off must be evaluated explicitly in any EMP/HPM vulnerability assessment and must be addressed in the design of replacement or upgraded I&C systems.

The engineering response to the EMP/HPM threat addresses three elements. First, electromagnetic hardening of safety-critical I&C: safety-critical digital systems shall be evaluated for EMP and HPM vulnerability, and hardening measures — including shielding enclosures, surge protection, fiber-optic signal isolation, and physical separation of redundant channels — shall be applied to meet the threat characterization for the facility's geopolitical environment. The IEC 61000 series on electromagnetic compatibility provides the baseline standard for this evaluation, to be supplemented by military-derived EMP hardening criteria appropriate to the threat level. Second, analog backup capability: each facility shall maintain analog backup instrumentation and, where practicable, analog or relay-based backup protection capability for the minimum set of safety functions required to monitor and maintain the Autonomous Safe Shutdown Condition (ASSC, Section 6). This backup capability must be capable of performing its safety function independent of the digital systems that may be degraded or destroyed by EMP or HPM effects. Third, post-event recovery: EMP and HPM strikes may not produce immediately observable physical damage; the effects manifest as silent failure of electronics. The ASSC maintenance procedures shall include provisions for diagnosing and responding to unexplained I&C failures consistent with EMP or HPM effects, including the transition to analog backup operation and the assessment of which safety functions remain available.

The EMP/HPM threat category does not replace or supersede any of the other five threat categories identified in Section 2.5. It addresses a distinct mechanism of effect that operates against a different class of vulnerabilities. A facility that has hardened its physical structures against explosive ordnance, its digital systems against cyber intrusion, and its operations against staffing degradation may nonetheless remain vulnerable to EMP or HPM if its I&C systems are not electromagnetic-hardened. Digital instrumentation and control modernization programmes that replace relay-based protection systems at operating plants therefore alter those plants' electromagnetic vulnerability profiles and must be assessed under this threat category; the Finland national implementation addresses one such case.

6. The Autonomous Safe Shutdown Condition

The military action design basis introduces a new plant state, distinct from both normal operation and existing post-accident conditions, namely the Autonomous Safe Shutdown Condition (ASSC). The ASSC is defined as a stable, passively maintained plant state in which the reactor is shut down, decay heat is being removed by passive or minimally attended means, and the facility can remain in this condition indefinitely without external support, active intervention by off-site organizations, or the assumption that normal operating conditions will be restored within any defined period.

In the IAEA's plant state taxonomy, the ASSC does not map directly to any existing defined state. It shares characteristics with the safe shutdown state. Still, it differs in a critical respect: existing safe shutdown provisions assume the state is transient—a condition on the path to either restart or controlled maintenance —implying that external support is available and that normal conditions will be restored. The ASSC makes no such assumption. It is designed to be the end state, sustainable for an indefinite period, requiring only the resources available at the facility and the actions of whatever staff remain.

The ASSC concept draws on and extends several existing provisions. The station blackout coping capability required by many national regulations provides the nearest engineering analogy, but with the duration extended from hours to an indefinite period and the scope expanded from loss of AC power to loss of all external support functions. The passive safety features incorporated into Generation III+ and Generation IV reactor designs—passive core cooling, passive containment cooling, gravity-driven safety injection—are directly relevant to achieving the ASSC. For existing plants, achieving ASSC capability may require modifications to extend coping durations, increase passive safety margins, and reduce dependence on active systems requiring external inputs.

The transition to the ASSC is a deliberate operational action, not a consequence of accident progression. It is initiated in advance of, or in response to, military threat conditions. It involves a controlled reactor shutdown followed by a transition to a passively stable configuration before the military action degrades the facility's ability to perform a controlled shutdown. The tiered

trigger framework described in Section 7 defines the conditions under which this transition should be initiated.

7. The Tiered Trigger Framework: General Architecture

The military action design basis requires a trigger framework that defines the conditions under which protective actions are initiated. Unlike existing design-basis events, which are triggered by physical conditions at the plant (loss of coolant flow, seismic acceleration, loss of off-site power), military action triggers are based on external threat assessments from military and intelligence organizations.

This section establishes the general architecture of the tiered trigger framework. The specific institutional mechanisms, authority chains, and coordination protocols through which this framework is implemented will differ by national jurisdiction and are addressed in the national implementation documents.

7.1 Tier 1: Enhanced Vigilance

Tier 1 is activated when military threat assessments indicate an elevated but non-imminent risk to nuclear facilities within a member state's territory or a defined geographic region. The trigger criteria are based on geopolitical intelligence indicators: massing of hostile military forces within striking distance, escalation of a regional conflict to a level that could involve nuclear facility host nations, or explicit threats against nuclear infrastructure.

Tier 1 actions are preparatory and do not affect plant operations. They include activating enhanced communication protocols among the national nuclear regulator, the national military authority, and the facility operator; verifying emergency procedures and supplies; reviewing and validating the facility's ASSC transition procedures; and notifying relevant international organizations, including the IAEA.

7.2 Tier 2: Operational Posture Change

Tier 2 is activated when military threat assessments indicate a credible and proximate risk to specific nuclear facilities. The trigger criteria are based on operational military indicators: hostile military operations are underway in the region and advancing toward the facility, or military intelligence assesses a specific facility as a likely objective or a collateral risk in ongoing or anticipated operations.

Tier 2 actions affect plant operations but do not require a shutdown. They include controlled power reduction to lower the decay heat inventory, pre-positioning of consumables and emergency equipment, staffing augmentation for key safety positions, activation of alternative communication pathways, and preparation for a rapid transition to the ASSC. The operational objective at Tier 2 is to reduce the consequences and increase the margin of safety should Tier 3

be required, while maintaining the option to return to normal operations if the threat does not materialize.

7.3 Tier 3: Mandatory Shutdown and ASSC Transition

Tier 3 is activated when military action against a nuclear facility is assessed as imminent or has commenced. The trigger criteria are based on direct threat indicators: hostile military forces are within a defined proximity of the facility, military operations are actively targeting infrastructure in the facility's area, or the facility itself has been directly threatened or attacked.

Tier 3 requires a mandatory reactor shutdown and a transition to the Autonomous Safe Shutdown Condition. This is a protective action analogous to a reactor trip on loss-of-coolant, but triggered by a military threat assessment rather than by a plant process parameter. Once initiated, the ASSC transition follows defined procedures to bring the plant to the passively stable, indefinitely sustainable condition described in Section 6.

7.4 National Implementation Requirements

Each member state implementing this framework must establish the institutional mechanisms to operationalize the tiered triggers. At minimum, this requires defined information pathways from national military and intelligence authorities to the national nuclear regulator and facility operators; a clear decision authority for each tier, specifying which official or body can activate the tier and under what conditions; legal and regulatory provisions that authorize and protect the decisions taken under each tier, including the authority to order preemptive shutdown based on military threat assessment rather than observed plant conditions; and coordination mechanisms with collective security organizations (NATO, EU, or relevant regional security frameworks) for member states whose military threat assessments are informed by alliance intelligence.

These national decision authorities and the plant’s own authority described in Section 7.5 are complementary, and they combine in one direction only. The plant’s standing authority to begin the transition to the ASSC on its own observable indicators is a floor: the plant may always move toward a safer state without external authorization. National authority may order a tier change, but its order is never a precondition for the plant to protect itself. This is the principle of local sovereignty (Section 5.1) applied to decisions as well as to safety functions. Where any part of the decision chain is supported by automated or AI-enabled systems, the governance of those systems, including which actions they may take and which require human authorization and by whom, is addressed in the nuclear annex to the Command Broker series, CB-Annex-Nuclear.

The specific design of these mechanisms will reflect each member state's constitutional arrangements, regulatory structure, military command hierarchy, and international security commitments. The national implementation documents provide worked examples for specific jurisdictions.

7.5 Uncrewed Aerial Systems as Trigger Observables

The Russia–Ukraine conflict has demonstrated that uncrewed aerial systems—encompassing one-way attack drones, first-person-view strike platforms, reconnaissance UAS, and autonomous AI-enabled systems—have become a principal instrument of military operations against energy infrastructure. UAS activity provides observable indicators at each tier of the trigger framework, supplementing the conventional force indicators described above. At Tier 1, relevant UAS indicators include intelligence reporting of adversary acquisition or forward deployment of attack drones in quantities consistent with an infrastructure campaign, and evidence of adversary investment in AI-enabled drone capabilities. At Tier 2, relevant indicators include confirmed UAS strikes against energy infrastructure within the host nation, UAS reconnaissance activity over or near the facility, and evidence of adversary UAS staging within range of the facility. At Tier 3, relevant indicators include UAS strikes against infrastructure in the facility's region, detection of surveillance UAS directly over the facility, and UAS attacks on the facility's off-site power infrastructure.

The ability to observe UAS activity as a trigger indicator depends on the facility's ability to detect UAS in its operational environment. This creates a prerequisite capability: each facility implementing the trigger framework must maintain a UAS detection and characterization capability as a standing element of its monitoring posture. A multi-sensor approach—integrating radar, acoustic, radio-frequency, and electro-optical sensors—provides the most robust detection capability. As AI-enabled drones increasingly operate autonomously without detectable control-link emissions, radar and electro-optical detection become the primary modalities. The companion instruments DBA-UxS and DBA-MA-UxS-NA detail the considerations for detection technology, monitoring guidance, and counter-UAS coordination requirements.

Emerging detection technologies offer additional capability pathways that should inform each national implementation's technology evaluation. AI-enabled visual recognition systems—including smartphone-based applications capable of distinguishing drones from other airborne objects in both day and night conditions using electro-optical infrared sensors—have been developed and deployed for military airbase protection. When integrated with ground-based LIDAR, such systems can provide all-weather, three-dimensional, real-time UAS tracking through conditions that degrade conventional electro-optical sensors. These low-cost, rapidly deployable detection layers can supplement fixed multi-sensor arrays and are particularly relevant for extending detection coverage during the standing preparedness posture required at Tier 1. The companion UAS addendum provides detailed guidance on the evaluation and integration of these technologies for nuclear facility applications.

The speed at which UAS threats can develop has implications for the timing of the transition. A UAS strike campaign can be initiated with very short warning time—one-way attack drones can reach targets within hours of launch, and forward-positioned units can deploy FPV drones with minimal preparation. This compressed timeline reinforces the requirement that plant management possess pre-authorized standing authority to initiate the ASSC transition based on

observable indicators, without waiting for external authorization, as the time available for decision-making under UAS threat conditions may be measured in hours rather than days.

8. Relationship to the Existing Safety Standards Architecture

8.1 Placement Within the IAEA Standards Hierarchy

The military action design basis framework intersects two streams of IAEA standards: the Safety Standards Series and the Nuclear Security Series. This document argues for primary placement within the Safety Standards framework, as articulated in Section 3.3: the engineering response to military action is fundamentally a safety design problem that addresses the conditions created by the threat, not the threat itself.

Within the Safety Standards hierarchy, the military action framework most naturally maps to a revision or supplement to SSR-2/1, establishing military action as a new category of design extension condition. The DEC framework, introduced following Fukushima, already provides the conceptual and regulatory architecture for addressing scenarios beyond the traditional design basis. Military action represents a category of DEC in which the initiating conditions are deliberately caused rather than arising from natural phenomena or equipment failure. Still, the engineering analysis—identifying the safety functions challenged, the conditions imposed, and the design features required to maintain safety—follows the same analytical methodology.

The name of the series and its placement are deliberate and do not conflict. The series is named for the design basis accident because it adopts the discipline of design basis analysis: a defined set of conditions to be withstood, identified structures, systems and components credited in the analysis, and explicit acceptance criteria, with the ASSC as the safety endpoint. Its proposed placement in the IAEA hierarchy is as a category of design extension condition, because military action lies outside the frequency-based envelope of postulated initiating events on which the design basis proper is built. The analytical method is that of the design basis; the regulatory home is the design extension condition. Where a national framework has no design extension category, the national implementation document states how the same analysis is carried.

The threat characterization inputs—the definition of the military threat against which the facility must be designed to survive—would draw on the Nuclear Security Series methodology for defining threat levels, adapted from the sub-national DBT framework to the nation-state military threat context. This creates a bridge between the two standard streams without collapsing them: security methodology informs the threat definition; safety methodology governs the design response.

8.2 Relationship to Existing Design Basis Events

The military action design basis does not replace any existing design-basis event or design-extension condition. Rather, it represents an envelope case that encompasses multiple concurrent

events, with the additional conditions of intentional causation, indefinite duration, and the loss of all external support. A military action scenario may simultaneously produce conditions equivalent to a station blackout, loss of the ultimate heat sink, loss of coolant due to physical damage, fire from explosive munitions, and loss of normal staffing—all occurring at once and persisting indefinitely.

This envelope characteristic means that design features developed to address the military action DEC will also strengthen the plant's resilience against the individual events already in the design basis. Enhanced passive cooling capability, extended on-site consumable storage, and automated safety functions developed for military action resilience provide additional margin against conventional accidents. This is not an ancillary benefit but a fundamental feature of the framework: addressing the most severe postulated conditions simultaneously improves the safety margin for all lesser conditions.

8.3 Application to New Builds Versus Existing Plants

The military action framework has significantly different implementation implications for new reactor construction versus existing operating plants. For new builds—including the Generation III+ designs currently under construction or in licensing in multiple member states, and the advanced reactor designs in earlier stages of development—incorporating military action resilience into the design basis from the outset is substantially more practicable and cost-effective than retrofitting. Several Generation III+ and Generation IV designs already incorporate passive safety features that contribute directly to military action resilience, including passive core cooling, passive containment cooling, and extended coping times for station blackout conditions.

For existing plants, the framework should be applied through a graded approach consistent with IAEA guidance on the application of safety standards to existing facilities. Periodic safety reviews (PSRs) provide the established mechanism for evaluating existing plants against evolving safety standards. The military action framework would be incorporated into the PSR process, with the scope and pace of any required modifications reflecting the specific plant design, its geographic and geopolitical risk profile, and the remaining licensed operating life. Some elements of the framework—particularly the ASSC transition procedures, the trigger framework coordination, and enhanced consumable storage—can be implemented through operational and procedural changes without major physical modifications.

8.4 UAS Threat Evolution and Design Basis Currency

The military action threat profile includes a dimension that distinguishes it from other external hazards addressed by the safety standards: the rapid evolution of uncrewed aerial systems. Natural phenomena change on geological or climatological timescales. The UAS threat evolves over months to years, driven by technological development, battlefield experience, and the competitive dynamics of military innovation. The Russia–Ukraine conflict has demonstrated that

AI-enabled drone capabilities—autonomous target recognition, jamming resistance, swarm coordination—can advance qualitatively within a single campaign cycle. The decadal periodic safety review, while appropriate for natural hazards and most design-basis events, is insufficient for maintaining the currency of UAS threat characterizations. An intermediate review mechanism, comparable to the periodic update of the Design Basis Threat in the security framework, should be established for the UAS-specific elements of the military action design basis. The companion instruments DBA-UxS and DBA-MA-UxS-NA elaborate on this requirement.

9. Classification and Information Sharing Considerations

The development of a military action design basis necessarily involves information that may be sensitive from a national security perspective. Specific threat characterizations, vulnerability assessments, and detailed design countermeasures may warrant classification or restricted distribution under national security frameworks. This creates tension with the fundamental principle of nuclear safety regulation, which, in most member states and in the IAEA's own practice, relies on transparency, public documentation, and independent review.

This document proposes a layered approach to classification. The general engineering principles and safety objectives of the framework—the material presented in this document—should remain unclassified and publicly available, consistent with the IAEA's standard publication practice. The specific threat characterizations for individual member states or facilities, the detailed vulnerability assessments, and the classified elements of national trigger framework coordination mechanisms would be handled through existing national classification systems and international information-sharing agreements.

For member states participating in collective security organizations, the classification challenge has an additional dimension. NATO classification protocols, for example, provide a framework for sharing threat assessments among Alliance members, but extending this to the nuclear regulatory authorities of member states—which may not have routine access to NATO-classified material—requires deliberate institutional arrangements. The national implementation documents address these arrangements for specific jurisdictions.

10. National Implementation: The Child Document Architecture

This framework is designed to be implemented through national child documents, which translate international principles into specific regulatory, institutional, and operational provisions within each member state's context. Each child document inherits, by reference, the engineering principles, the ASSC concept, and the general trigger architecture from this parent document, and develops three nation-specific elements.

First, the regulatory mapping: how the military action design basis aligns with the national regulatory framework—which specific regulations, safety analysis requirements, and licensing provisions are affected —and what regulatory actions (rulemaking, regulatory guidance, technical positions) are needed to implement the framework. Second, the authority and coordination framework: the specific institutional mechanisms through which the tiered triggers are operationalized, including the information pathways, decision authorities, legal protections, and interagency coordination protocols specific to that member state's governmental structure. Third, the implementation pathway: how the framework is applied to the existing fleet and to new builds within that jurisdiction, including the graded approach for existing plants and the incorporation of the framework into design certification for new designs.

The first national child document in the DBA-MA series addresses the United States, mapping the framework to the NRC regulatory structure, the existing provisions of 10 CFR Part 50, the General Design Criteria, and the U.S. interagency coordination architecture involving the NRC, Department of Energy, Department of Defense, Department of Homeland Security, and the Intelligence Community.

Subsequent child documents are anticipated for European jurisdictions with particular relevance to the military action threat. Priority jurisdictions include France, given its large operating fleet and its status as both a NATO member and a nuclear weapon state; Finland, given its geographic proximity to Russia and its recent accession to NATO; the United Kingdom, given its nuclear fleet, its independent post-Brexit regulatory framework, and its NATO membership; and Sweden, given its proximity to the Baltic security environment and its recent NATO accession. Additional national implementations may be developed for other member states as the framework matures.

The DBA-MA series also includes three IAEA-level analytical instruments that apply across all national implementations: DBA-MA L2, the formal accident description and design requirements; DBA-MA L3, the tiered trigger framework and decision criteria; and DBA-MA L4, the monitoring sources and data-sharing guidance. These three documents are structured as children of this framework and carry IAEA-level scope. Each national child document references them by their IAEA designations; national-specific regulatory mappings and program details are addressed in standalone national child documents for each jurisdiction.

In addition to these analytical instruments, the DBA-MA series includes two dedicated instruments addressing uncrewed systems: DBA-UxS, the cross-sector foundational document, and DBA-MA-UxS-NA, the nuclear implementation annex. Each national child document should address the UxS-specific dimensions identified in those instruments, including: a national UAS threat characterization reflecting the adversary capabilities relevant to that jurisdiction; the legal authority for UAS detection and counter-UAS measures at nuclear facilities, which varies significantly across jurisdictions; pre-coordinated agreements for the deployment of military or law enforcement counter-UAS capabilities to nuclear facilities under the tiered trigger

framework; and the supply chain provisions for counter-UAS equipment and detection systems as a component of the facility's standing preparedness posture.

The series has since grown to include cross-cutting instruments on source term under military action (DBA-MA-ST), geopolitical siting (DBA-MA-SITE), defense in depth (DBA-MA-DiD) and physical proximity attack (DBA-MA-PPA), shared modules for applicability (DBA-MA-AF) and communications (DBA-MA-CDB), and facility-class instruments beginning with small modular reactors (DBA-MA-SMR-FC1). This document cites every instrument in the series by designation only. The current version of each is given in its own published record, so that a revision to a child never requires a revision here.

11. Next Steps

The formal engineering requirements for the military action design extension condition have been developed in DBA-MA L2, the Formal Accident Description and Design Requirements. That document provides the event description, the characterization of conditions to be withstood, the ASSC definition and acceptance criteria, and the seven engineering principles set out in Section 5.

The tiered trigger framework has been elaborated in DBA-MA L3, the Trigger Framework for Military Action Design Basis Accident. That document provides the three-tier observable indicator structure, required actions at each tier, and the general decision authority framework for national implementation. Implementation guidance for the monitoring sources and data-sharing agreements required to observe the trigger indicators is provided in DBA-MA L4, the Monitoring Sources and Data-Sharing Guidance. Both L3 and L4 carry IAEA-level scope; national-specific program details are addressed in standalone national child documents for each jurisdiction.

Engagement with the IAEA Secretariat will be pursued to determine the appropriate standards publication pathway for this framework—whether as a proposed revision to SSR-2/1, a new Safety Guide, a TECDOC, or an initial Nuclear Energy Series publication that could serve as the basis for eventual incorporation into the safety standards. The engagement strategy recognizes that the IAEA standards development process is consensus-based and deliberate, and that the political dimensions of the military action topic require careful navigation within the IAEA's membership structure.

The national child documents for the United States and Finland are complete in draft form and available for review alongside this framework document. Development of additional European child documents will proceed in parallel with the international framework, beginning with outreach to national regulatory authorities and technical support organizations in the priority jurisdictions identified in Section 10.

Two dedicated instruments address uncrewed systems threats: DBA-UxS, the cross-sector foundational document, and DBA-MA-UxS-NA, the nuclear implementation annex. Together

they translate the operational lessons of the sustained drone campaign against Ukrainian energy infrastructure into engineering design requirements, trigger framework observables, monitoring and detection guidance, and counter-UAS coordination provisions. National implementation documents should incorporate the provisions of both instruments into their regulatory mapping, threat characterization, and coordination frameworks. The rapid evolution of uncrewed system capabilities — particularly the integration of artificial intelligence into autonomous drone operations — will require that the UxS-specific elements of the framework be reviewed more frequently than the decadal periodic safety review, as discussed in Section 8.4 of this document.

This document is offered to the international nuclear safety and security community to build the analytical and institutional case for formal consideration of military action within the IAEA safety standards framework.

Conferred Reference Events

On the author's determination of 14 October 2026 (*Energy and Nuclear Reference Events Follow the Same Method*, held in Project 0), this framework acts as the nuclear sector framework for reference events and confers them by the same method as every other sector framework in the corpus. A facility class of the nuclear series proposes a real event; this section confers it and assigns a designator of the form DBA-NUC-RE-N, numbered in the order of conferral; Project 0 enters it in the DBA Reference Event Register; and Project 11, DBA Reference Events, writes the instrument that reads the event back through the proposing class's design basis and through this framework.

The series has one facility class today, *SMR Facility Class Design Basis* (DBA-MA-SMR-FC1), and it proposes the events it can read. Events at large light-water reactors, among them the seizure of the Chornobyl site and the seizure and occupation of the Zaporizhzhia Nuclear Power Plant described in Section 2, have no class to propose them. Until a large light-water reactor facility class issues, this framework proposes those events as well as conferring them, and says so where it does. When that class issues, it adopts them, and each instrument re-reads its event against the class at its next revision without changing its designator, findings or recommendations.

An event already conferred in another sector is cited here by its designator and is not conferred again. Among them is DBA-DAMS-RE-1 (*Destruction of the Kakhovka Dam*), whose reading records the loss of the Kakhovka Reservoir as the cooling-water source of the Zaporizhzhia plant.

On the author's decision of 15 October 2026, this framework confers its first two events, numbered in the order of conferral. The number records conferral, not issue, and the instruments may be written in either order. Each designator is sector-primary and single. The primary sector of both events is Nuclear Reactors, Materials and Waste, and where an event reached another sector, its reach is recorded in the register's impacted-sector column rather than in a second

designator. The tier reading given with each event is this framework's reading at conferral. The tier determination is the instrument's, made against the four military-action characteristics of intentionality, persistence, precision and denial, and the military-action tier holds only when all four are established.

DBA-NUC-RE-1 is the seizure and occupation of the Zaporizhzhia Nuclear Power Plant, Ukraine, from 4 March 2022 onward. This framework proposes the event as well as conferring it. It does so in place of a large light-water reactor facility class, which is owed and has not issued, because Zaporizhzhia is a six-unit VVER-1000 plant that no class of the series yet describes. When that class issues, it adopts the event, and the instrument re-reads it against the class at its next revision without changing its designator, findings or recommendations. Until then the instrument reads the event against this framework, Sections 2.2, 2.3 and 2.5 and the engineering principles of Section 5 in particular, and against *SMR Facility Class Design Basis* (DBA-MA-SMR-FC1) only where one of that class's provisions applies. The event realizes the first and fifth threat categories of Section 2.5, seizure and occupation and the denial of external support, with bombardment and attack by uncrewed aerial systems on the site at intervals. The impacted sectors this framework reads are Energy, for the loss of the plant's generation to the Ukrainian grid and the repeated loss of its off-site power, and Communications, for the severing of the plant's communications with the national regulator and the IAEA that Section 2.2 records. The loss of the Kakhovka Reservoir as the plant's cooling-water source is the reading of DBA-DAMS-RE-1 (*Destruction of the Kakhovka Dam*) and is cited from there, not conferred again. The tier reading at conferral is that the event is a candidate for the military-action tier. Intentionality, persistence and denial stand on the IAEA's record of the seizure and of the occupation that followed it. Precision is for the instrument to establish under the fixed attribution posture, since the IAEA has not attributed the strikes on the site, and until it is established the military-action tier is not held.

DBA-NUC-RE-2 is the accident at the Fukushima Daiichi Nuclear Power Station, Japan, beginning 11 March 2011. *SMR Facility Class Design Basis* (DBA-MA-SMR-FC1) proposes it at its Section 8.3, from version 0.11, for coping duration. That class anchors its 72-hour floor for the Autonomous Safe Shutdown Condition to the station-blackout and FLEX coping benchmark the accident produced, and reads its multi-module common-cause events, the loss of a shared switchyard and of a shared cooling-water supply, against an accident in which several units on one site lost their power and their heat sink together. This framework confers it. The instrument reads the event against the class and against Section 5 of this framework, extended autonomous operation (Section 5.2) and degraded-staffing operability (Section 5.4) in particular. The impacted sectors this framework reads are Energy, for the loss of the station's generation and the rotating outages that followed in its operator's service area; Food and Agriculture and Water, for the restrictions placed on food and on drinking water after the release; and Healthcare and Public Health, for the evacuation of hospitals and care homes from the evacuation zones. The tier reading at conferral is that the event does not hold the military-action tier and is not adversarial. It was initiated by an earthquake and tsunami, and none of the four characteristics is present. It is

conferred for what it measures: how long a plant must cope when every external support is lost at once and access to the site is impeded, which is the condition Section 2.3 says military action creates deliberately and sustains.

Neither instrument is yet written. Both are Project 11’s to write under the Reference Event Compression Standard, and the register rows are Project 0’s to enter. The events Section 2 describes remain the operational evidence on which this framework rests. The Chornobyl seizure of Section 2.1 and the 2014 overflights of French nuclear plants by uncrewed aircraft remain candidates and are not proposed at this version.

12. References

12.1 IAEA Safety Standards

INTERNATIONAL ATOMIC ENERGY AGENCY, Fundamental Safety Principles, IAEA Safety Standards Series No. SF-1, IAEA, Vienna (2006).

INTERNATIONAL ATOMIC ENERGY AGENCY, Safety of Nuclear Power Plants: Design, IAEA Safety Standards Series No. SSR-2/1 (Rev. 1), IAEA, Vienna (2016).

INTERNATIONAL ATOMIC ENERGY AGENCY, Safety of Nuclear Power Plants: Commissioning and Operation, IAEA Safety Standards Series No. SSR-2/2 (Rev. 1), IAEA, Vienna (2016).

INTERNATIONAL ATOMIC ENERGY AGENCY, Deterministic Safety Analysis for Nuclear Power Plants, IAEA Safety Standards Series No. SSG-2 (Rev. 1), IAEA, Vienna (2019).

INTERNATIONAL ATOMIC ENERGY AGENCY, Safety of Nuclear Fuel Cycle Facilities, IAEA Safety Standards Series No. SSR-4, IAEA, Vienna (2017).

INTERNATIONAL ATOMIC ENERGY AGENCY, Site Evaluation for Nuclear Installations, IAEA Safety Standards Series No. SSR-1, IAEA, Vienna (2019).

INTERNATIONAL ATOMIC ENERGY AGENCY, Design of Nuclear Installations Against External Events Excluding Earthquakes, IAEA Safety Standards Series No. SSG-68, IAEA, Vienna (2021).

INTERNATIONAL ATOMIC ENERGY AGENCY, Periodic Safety Review for Nuclear Power Plants, IAEA Safety Standards Series No. SSG-25, IAEA, Vienna (2013).

INTERNATIONAL ATOMIC ENERGY AGENCY, Preparedness and Response for a Nuclear or Radiological Emergency, IAEA Safety Standards Series No. GSR Part 7, IAEA, Vienna (2015).

INTERNATIONAL ATOMIC ENERGY AGENCY, Defence in Depth in Nuclear Safety, INSAG-10, IAEA, Vienna (1996).

12.2 IAEA Nuclear Security Series

INTERNATIONAL ATOMIC ENERGY AGENCY, Nuclear Security Recommendations on Physical Protection of Nuclear Material and Nuclear Facilities (INFCIRC/225/Revision 5), IAEA Nuclear Security Series No. 13, IAEA, Vienna (2011).

INTERNATIONAL ATOMIC ENERGY AGENCY, Engineering Safety Aspects of the Protection of Nuclear Power Plants Against Sabotage, IAEA Nuclear Security Series No. 4, IAEA, Vienna (2007).

INTERNATIONAL ATOMIC ENERGY AGENCY, National Nuclear Security Threat Assessment, Design Basis Threats and Representative Threat Statements, IAEA Nuclear Security Series No. 10-G (Rev. 1), IAEA, Vienna (2021).

INTERNATIONAL ATOMIC ENERGY AGENCY, Objective and Essential Elements of a State's Nuclear Security Regime, IAEA Nuclear Security Series No. 20, IAEA, Vienna (2013).

INTERNATIONAL ATOMIC ENERGY AGENCY, Computer Security for Nuclear Security, IAEA Nuclear Security Series No. 42-G, IAEA, Vienna (2021).

12.3 IAEA Reports on the Safety Situation in Ukraine

INTERNATIONAL ATOMIC ENERGY AGENCY, Nuclear Safety, Security and Safeguards in Ukraine: 2nd Summary Report by the Director General, 28 April – 5 September 2022, IAEA, Vienna (September 2022).

INTERNATIONAL ATOMIC ENERGY AGENCY, Director General updates on the safety, security, and safeguards situation in Ukraine, continuous series (March 2022–present). Available at: https://www.iaea.org/resources/databases/ukraine-updates

INTERNATIONAL ATOMIC ENERGY AGENCY, The Seven Indispensable Pillars of Nuclear Safety and Security (articulated by Director General Grossi in the context of the Zaporizhzhia Nuclear Power Plant situation, 2022). Referenced in successive Director General updates and Board of Governors reports.

INTERNATIONAL ATOMIC ENERGY AGENCY, IAEA Support and Assistance Mission to Zaporizhzhia (ISAMZ), Mission Reports (September 2022–present).

12.4 DBA-MA Document Series

DBA-MA L1 (United States) — Design Basis Expansion for Nuclear Power Plants Considering Military Action. National Implementation: United States. Eclectic Technologies.

DBA-MA L1 (Finland) — Design Basis Expansion for Nuclear Power Plants Considering Military Action. National Implementation: Finland. Eclectic Technologies.

DBA-MA L2 — Design Basis Accident: Military Action Against Nuclear Facilities. Formal Accident Description and Design Requirements. Eclectic Technologies.

DBA-MA L3 — Trigger Framework for Military Action Design Basis Accident: Observable Indicators and Decision Criteria for Transition to Autonomous Safe Shutdown Condition. Eclectic Technologies.

DBA-MA L4 — Monitoring Sources and Data-Sharing Guidance for DBA-MA Trigger Observables. Eclectic Technologies.

DBA-UxS — Uncrewed Systems: Threat Considerations for Critical Infrastructure. Eclectic Technologies.

DBA-MA-UxS-NA — Uncrewed Systems: Nuclear Sector Implementation Annex. Eclectic Technologies.

DBA-MA-SITE — Geopolitical Siting Assessment Methodology. Eclectic Technologies.

CB-Annex-Nuclear — Command Broker: Nuclear Sector Annex. Application of the Cross-Sector Command Broker Requirement to Nuclear Facilities. Eclectic Technologies.

DBA-NUC-RE-1 — Seizure and Occupation of the Zaporizhzhia Nuclear Power Plant, 2022 onward. Reference event conferred in Conferred Reference Events, at the close of Section 11. Instrument owed, to be written by Project 11, DBA Reference Events. Eclectic Technologies.

DBA-NUC-RE-2 — The Fukushima Daiichi Accident, 2011. Reference event proposed by DBA-MA-SMR-FC1 and conferred in Conferred Reference Events, at the close of Section 11. Instrument owed, to be written by Project 11, DBA Reference Events. Eclectic Technologies.

12.5 Companion and Supporting Analyses

ROXEY, T., Strengthening Ukraine's Electric Grid Against Russian Cyber and Kinetic Attack: Recommendations for Grid Hardening, Decentralized Topology, Drone Capacity Building and Cyber Army Coordination, Version 1, Eclectic Technologies (March 2025).

ROXEY, T., Analysis of Ukraine Grid Defense Experience (Chapter for NPECA), Restructured Version 16, Eclectic Technologies (March 2025).

12.6 National Regulation Cited in the Text

UNITED STATES NUCLEAR REGULATORY COMMISSION, General Design Criteria for Nuclear Power Plants, 10 CFR Part 50, Appendix A.

12.7 International Law, Statements and Open-Source Reporting

Protocol Additional to the Geneva Conventions of 12 August 1949, and relating to the Protection of Victims of International Armed Conflicts (Protocol I), 8 June 1977, Article 56, Protection of Works and Installations Containing Dangerous Forces.

UNITED NATIONS, “IAEA Chief Outlines Five Principles to Avert Nuclear ‘Catastrophe’ in Ukraine,” UN News (30 May 2023).

BROWN, S., “Artificial Intelligence Raises Ukrainian Drone Kill Rates to 80%,” Kyiv Post (14 October 2024).

[Note: References to classified threat assessments, site-specific vulnerability analyses, Safeguards-protected information, and NATO-classified defence planning documents are not listed here. Such references are maintained in controlled annexes to the relevant national implementation documents and are accessible through appropriate classification channels.]

Revision History

Revision notes for versions 1.0 through 1.4 were not carried forward in the drafts and are not reconstructed here.

Revision Note (v1.5, April 2026): US child document version reference updated from v2.4 to v2.5. All DBA-MA-UAS (v1.3) references updated to DBA-UxS (v1.0) and DBA-MA-UxS-NA (v1.0) pair throughout, including inline citations, companion addendum references, instrument list, and references section entry.

Revision Note (v1.6, April 2026): Section 3.4 added: The Geopolitical Stability Dimension of Siting Assessment. Establishes that DBA-MA introduces a geopolitical stability dimension to siting assessment not captured in existing NS-R-3/SSR-1 criteria. Addresses the SMR source-term-relaxation argument: source-term-based siting relaxations require independent DBA-MA threat category evaluation for the proposed site and may not be accepted as sufficient without it.

Revision Note (v1.7, June 2026): Section 5.7 added: Electromagnetic Pulse and High-Power Microwave. Establishes EMP and HPM as a sixth engineering design principle for military action resilience, addressing the electromagnetic threat mechanism distinct from kinetic and software-based categories. Covers EMP and HPM threat characterization, the analog/digital I&C asymmetry, three-element engineering response (electromagnetic hardening, analog backup capability, post-event recovery), and the Loviisa I&C modernization as a reference case. Section 5.6 consequential update: "five principles above" corrected to "six principles above."

Revision Note (v2.0, September 2026): Issued as the first final version. Removes the draft status and adds the citation block, stamped with the reserved version DOI 10.5281/zenodo.22944855. Adds a scope statement extending the principles by analogy to other nuclear installations (Section 1), the six threat categories as Section 2.5, the international humanitarian law context in Section 3.2, the treatment of cyber attack in the Section 5 introduction, the combination of plant and national shutdown authority with a pointer to CB-Annex-Nuclear in Section 7.4, and the reconciliation of the series name with its placement as a design extension condition in Section 8.1. Corrects the principle count in Section 5.6, which the version 1.7 note had changed in error (five principles precede Section 5.6, not six), and states

that Section 5 carries seven principles, all of which L2, Version 3.0, now carries. Anchors the seven-pillars finding to the Director General's second summary report, corrects the ZNPP capacity to approximately 5,700 MWe, sources the eighty-percent figure, qualifies automation in Section 5.4 as hard-wired or qualified deterministic, moves the Loviisa case to the Finland child, and adds a pointer to DBA-MA-SITE in Section 3.4. Series instruments are now cited by designation without version numbers. Corrects the SSG-68, SSG-25, Nuclear Security Series 10-G and September 2022 report references, removes the superseded NS-G-1.5, adds SSR-1, SSR-4, Additional Protocol I, the Director General's five principles and the Kyiv Post source, and removes references not cited in the text.

Revision Note (v2.1, October 2026): Conferred Reference Events added at the close of Section 11 under the author's determination of 14 October 2026, Energy and Nuclear Reference Events Follow the Same Method. This framework acts as the nuclear sector framework for conferral, under the designator form DBA-NUC-RE-N, and proposes events at large light-water reactors until a class for them issues. No event is yet conferred. The citation block gives v2.1, the footer returns to draft status, and the DOI line records that the DOI cited is v2.0's. No section is renumbered.

Revision Note (v2.2, October 2026): Conferred Reference Events confers its first two events, on the author's decision of 15 October 2026. DBA-NUC-RE-1, the seizure and occupation of the Zaporizhzhia Nuclear Power Plant (2022 onward), is proposed by this framework in place of the owed large light-water reactor facility class, which adopts it when it issues. DBA-NUC-RE-2, the Fukushima Daiichi accident (2011), is proposed by SMR Facility Class Design Basis (DBA-MA-SMR-FC1) at its Section 8.3, version 0.11, for coping duration. Each conferral states the proposer, the primary sector, Nuclear Reactors, Materials and Waste, the impacted sectors and the tier reading at conferral; the tier determination is left to each instrument, which Project 11 writes. Kakhovka stays cited as DBA-DAMS-RE-1 and is not conferred again. The sentence that no event is yet conferred is withdrawn. Section 12.4 adds the two designators. The cover, citation block, header and footer read v2.2. The DOI line still reads "DOI (v2.0)", because v2.1 was not deposited, and now says that a DOI for v2.2 is assigned when it is deposited. No section is renumbered and no other text changes.

Deposit Note (v2.2, 16 October 2026): The citation block is stamped with version DOI 10.5281/zenodo.23420400, reserved by the author on 16 October 2026 as a new version under concept DOI 10.5281/zenodo.22944854, superseding v2.0 (10.5281/zenodo.22944855). Version 2.1 was a working revision and was not deposited. No content change from v2.2 as issued.